

STEPUE

Proof of Concept (POC) | Ankle-Worn Gait Monitoring & FOG Condition Detection

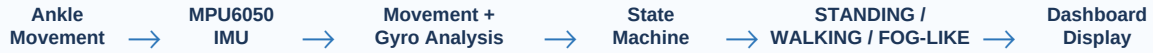
01 PROBLEM

Freezing of Gait (FOG) can involve difficulty initiating or continuing walking. This POC explores whether changes in ankle movement can be detected during controlled testing and displayed as a defined FOG condition.

02 CURRENT POC SOLUTION

STEPUE is an ankle-worn prototype built around an **ESP32**. An **MPU6050 IMU** measures acceleration and rotation, while an **FSR** provides pressure input through an **ADS1115 external ADC**. A heuristic state machine currently classifies the device as **STANDING**, **WALKING**, or **FOG-LIKE CONDITION**. The detected state is shown on a live dashboard. No motor or tactile cue actuator is currently being used in this POC.

03 WORKING PRINCIPLE



Pressure path: FSR → ADS1115 A0 → pressure level (NO / LOW / MEDIUM / HIGH) → dashboard. The current gait-state heuristic is driven by IMU movement and gyroscope features.

04 HARDWARE & SOFTWARE INTEGRATION

ESP32 — microcontroller and processing
MPU6050 — acceleration and gyroscope sensing
ADS1115 — external ADC for FSR input
FSR — pressure sensing input
Li-Po + TP4056 + boost — portable power path being tested
Ankle enclosure/strap + wiring — wearable mounting

Dashboard: a web page is hosted directly by the ESP32 and displays the current state, FSR values, movement, gyroscope, live graphs, IP address and uptime. During the present development setup, the ESP32 is programmed and powered through USB while the dashboard data is communicated over Wi-Fi. Battery-only operation and reliable Wi-Fi connection are still being tested.

05 CURRENT POC STATUS & OUTPUT

Working: MPU6050 + ADS1115 acquisition, ESP32 state-machine processing, Wi-Fi dashboard, and live sensor/state visualization.
Current tuning: movement sensitivity is being adjusted so controlled walking is more readily classified as WALKING, stopping returns toward STANDING, and more disturbed movement can reach the FOG-LIKE CONDITION state.
Still being calibrated/tested: FSR pressure response and battery-only power/Wi-Fi operation.

Input: continuous ankle IMU data + FSR data
Processing: ESP32 heuristic state classification
Output: STANDING / WALKING / FOG-LIKE CONDITION shown on dashboard

06 WHAT THE POC CURRENTLY DEMONSTRATES

- Ankle movement sensing using the MPU6050
- Classification of controlled movement into STANDING and WALKING states
- Detection of a predefined disturbed movement condition represented as FOG-LIKE CONDITION
- FSR pressure sensing and live pressure display, subject to calibration
- Sensor → processing → state → dashboard loop
- Local Wi-Fi dashboard communication when the ESP32 is connected to the same Wi-Fi network

NEXT DEVELOPMENT: FSR calibration → movement/state threshold tuning → battery-only power and Wi-Fi reliability → controlled gait testing → collection of real gait/FOG data → algorithm refinement → future cueing hardware, if required.

POC SCOPE: This prototype demonstrates technical feasibility of sensing ankle movement and classifying a predefined FOG-like/disturbed movement condition. It does **not** establish clinical FOG prediction accuracy, diagnosis, or prevention, and the current POC does not yet provide a motor/vibration cue.